Product Requirements Document

Backstage: Public Speaking Training App

Field	Details
Project No.	8
Project Name	Public Speaking Training App
Brand Name	Backstage
Motto	Practice loud. Perform proud.
Team	Frontend Group No. 7 (4 members) | Backend Group No. 8 (4 members) — Total 8 members
Phase	Development: backend built and tested, handed over to the frontend group
Prepared By	Backend Team – Group No. 8
Date	20 October 2026 (updated after the backend was built)

1. Problem Statement

Many people struggle to rehearse presentations effectively on their own. Without a live audience or coach, it is hard to know whether pacing is too fast or too slow, whether pauses are used well, and how often filler words ("um", "like", "you know") appear. Existing tools either focus purely on scripting content or provide generic recording with no structured feedback on delivery.

2. Objective

Build a presentation rehearsal tool that helps users practice speeches independently and receive objective, data-driven feedback on pacing, pause usage, and filler-word frequency — turning a solo rehearsal into something closer to a coached practice session.

3. Target Users

- Students preparing for presentations, vivas, or debates
- Early-career professionals preparing for interviews or pitches
- Anyone preparing for a talk, demo, or public speaking event who wants structured self-feedback

4. Research Summary

Reference material reviewed during the research phase: the Duolingo Product Handbook and eight public-speaking / communication talks.

4.1 Duolingo Product Handbook — Key Takeaways

The handbook lays out five operating principles the team can borrow for how this project is run, not just what it builds:

- Take the Long View — prioritize long-term user value over short-term wins; design features that build a habit rather than a one-time gimmick
- Raise the Bar — ship a polished "V1" rather than a rough MVP; every feature should be useful, intuitive, delightful, and polished before release
- Ship It — move fast, run small experiments, and use real usage data (not internal debate) to decide what to keep
- Show Don't Tell — lead with results and data (e.g., "users reduced filler words by X% after 3 sessions") rather than describing effort; keep updates short and evidence-based
- Make It Fun — engagement mechanics (streaks, encouraging feedback, playful tone) matter even for a "serious" skill-building tool

Applied to this project: favor a small, well-polished MVP (prompt library + timer + WPM + filler-word detection) over a broad but rough feature set, and consider a lightweight engagement mechanic (e.g., a practice streak) inspired by Duolingo's retention focus.

4.2 Public Speaking / Communication Talks — Per-Video Analysis

Each talk was analyzed for its core message, actionable techniques, and a possible app feature it inspires:

1. Malavika Varadan – How to Make Conversation with Anyone

Core message: Conversation is a learnable skill, not an innate talent — small, deliberate actions can start a good conversation with anyone.

Key techniques:

- Say the first word — take initiative to start
- Find a "Me Too" — a shared point of connection
- Pay a genuine compliment to break the ice
- Ask for the other person's opinion to build connection
- Maintain confident eye contact
- Remember and use people's names

App feature idea: Ice-Breaker Scenario Simulator — an AI character prompts the user to open a conversation within ~30 seconds using a "Me Too" style hook.
Named technique/framework: *The "7 Ways to Make a Conversation" framework*

2. Julian Treasure – How to Speak So That People Want to Listen

Core message: People listen when a speaker avoids negativity and speaks with honesty, authenticity, and vocal variety.

Key techniques:

- Avoid the 7 deadly sins of speaking: gossip, judging, negativity, complaining, excuses, exaggeration, dogmatism
- Follow the HAIL principle: Honesty, Authenticity, Integrity, Love
- Speak from the chest register for a deeper, more trustworthy tone
- Vary pacing and use dramatic pauses
- Add prosody/tonal variety instead of a monotone delivery

App feature idea: Vocal Variety & Pause Analyzer — analyzes pitch variation, pacing (WPM), and silence/pauses from the recording; flags monotone delivery.
Named technique/framework: *The HAIL method; vocal warm-ups (sirens & lip trills)*

3. Frederik Imbo – How to Not Take Things Personally

Core message: Someone else's negative behavior usually reflects their own state of mind, not a personal failure on your part.

Key techniques:

- Reframe as "it's about them, not me" when facing hostility
- Look for the other person's underlying motive (empathy)
- Question your own ego/insecurity when a comment stings
- Know and hold on to your own sense of value
- Be vulnerable without assigning blame

App feature idea: Difficult Q&A / Heckler-Handling Module — analyzes whether the user responds to hostile or difficult audience questions in a calm, neutral tone.
Named technique/framework: *The "2-Euro Coin" analogy*

4. Celeste Headlee – 10 Ways to Have a Better Conversation

Core message: Good communication starts with active listening — not assuming you already know what the other person will say.

Key techniques:

- Avoid multitasking/phone use; stay fully present
- Ask open-ended questions (who/what/when/where/why) instead of yes/no questions
- Go with the natural flow instead of over-rehearsing set lines
- Admit "I don't know" when true
- Don't equate your own experience with the other person's
- Listen to understand, not just to reply

App feature idea: Filler Word & Question-Type Detector — tracks how many open-ended questions the user asks and how often filler words (um, like, uh) occur.
Named technique/framework: *The journalist's / reporter's questions framework*

5. Andrew Tarvin – The Skill of Humor

Core message: Humor is a learnable, practicable skill — not innate talent — and it significantly boosts audience engagement.

Key techniques:

- Treat humor as a structure/formula that improves with practice
- Aim for a smile, not necessarily a big laugh
- Surprise is the key ingredient of a joke
- Observe everyday situations for material
- Use humor to support the message, not distract from it

App feature idea: Punchline Timing & Pause Tracker — checks whether the user pauses ~1–2 seconds before a punchline to improve comic timing.

Named technique/framework: *The "Humor Engineer" and "Yes, And" approach*

6. David JP Phillips – The 110 Techniques of Communication and Public Speaking

Core message: Great speakers aren't born that way — they apply specific, learnable physical and mental techniques consistently.

Key techniques:

- Step forward on key points (builds intimacy), step back afterward
- A slight head tilt conveys empathy
- Keep hand gestures inside an imaginary "box" between chest and waist
- Smile genuinely at the open/close of a talk (Duchenne smile)
- Avoid closed body language — crossed arms, hands in pockets

App feature idea: Body Language & Gesture Tracker (camera-based) — checks whether hands stay within "the box," flags excessive pacing, and scores eye contact.
Named technique/framework: *"The Box" technique; the Duchenne smile*

7. Lynne Franklin – Reading Minds Through Body Language

Core message: People process information as Visual, Auditory, or Kinesthetic (VAK); matching delivery style to the audience type improves how a message lands.

Key techniques:

- Visual audiences: speak faster, use images/visual examples
- Auditory audiences: prefer rhythmic tone; modulate pitch
- Kinesthetic audiences: respond to slower, emotion-led delivery
- "Match and mirror" the audience's body language

App feature idea: Audience-Type Pacing Mode — user rehearses against different audience profiles (V/A/K) and the app checks whether pacing (WPM) fits that profile.
Named technique/framework: *The VAK Model (Visual, Auditory, Kinesthetic)*

8. Jean Smith – The Science of Flirting

Core message: Building attraction/rapport relies on well-studied non-verbal signals — relevant to creating instant audience connection in public speaking.

Key techniques:

- Maintain eye contact
- Keep open body language (uncrossed arms/legs)
- Smile genuinely rather than performatively
- Give full attention to the listener
- Maintain appropriate physical proximity/distance

App feature idea: Facial Expression & Eye-Contact Heatmap — AI heatmap of how much the user looked at the "audience" and how often they smiled.
Named technique/framework: *The H.O.T. A.P.E. framework (Humor, Open body language, Touch, Attention, Proximity, Eye contact)*

Open gap identified: most reviewed resources give qualitative, human-coached advice on delivery, but none pair that advice with objective, measurable feedback (WPM, pause length,

filler-word count, vocal variety) generated automatically from a user's own recording. This is the core gap the app aims to fill.

5. Feature Ideas Backlog (from Research)

Feature ideas surfaced during research, grouped by whether they fit the current MVP scope (audio/transcript-based) or require future capability (e.g., camera/video analysis).

5.1 Aligned with Current MVP Scope

- Vocal Variety & Pause Analyzer (Julian Treasure) — pitch variation, pacing/WPM, pause detection [Built: pace and pauses. Not built: pitch variation]
- Filler Word & Question-Type Detector (Celeste Headlee) — filler-word count + open-ended question tracking [Built: filler words. Not built: open-ended question tracking]
- Punchline Timing & Pause Tracker (Andrew Tarvin) — pause duration before key lines [Not built as a feature. Every pause and its length is available in the results]
- Audience-Type Pacing Mode (Lynne Franklin) — WPM target adjusted by audience profile [Not built. One target pace range for everyone, admins can change it]

5.2 Future / Backlog (Camera or Advanced Analysis Required)

- Ice-Breaker Scenario Simulator (Malavika Varadan) — conversational AI prompt practice
- Difficult Q&A / Heckler-Handling Module (Frederik Imbo) — tone analysis under pressure
- Body Language & Gesture Tracker (David JP Phillips) — camera-based posture/gesture scoring
- Facial Expression & Eye-Contact Heatmap (Jean Smith) — camera-based engagement heatmap

6. Scope

6.1 In Scope (MVP)

- Speech prompt library (curated practice topics/prompts)
- Rehearsal timer with speech-to-text transcription
- Words-per-minute (WPM) pacing counter
- Filler-word detection and flagging, with a configurable word list
- Recording interface
- Transcript analysis view
- Post-speech performance breakdown

6.2 Out of Scope (Future Consideration)

- Video-based body language / non-verbal analysis
- Humor or audience-engagement scoring
- Live multi-user rehearsal / peer feedback

7. User Flow (Draft)

- 1. User selects or is given a speech prompt from the library
- 2. User starts recording; a timer runs while they speak
- 3. User finishes recording; the audio is uploaded and transcribed in the background (not live)
- 4. System analyzes transcript for WPM, pauses, and filler words
- 5. User views performance breakdown (pacing graph, filler-word count/timestamps, pause map)
- 6. User can re-attempt the same prompt to track improvement over time

8. Key Deliverables

- Recording interface (frontend)
- Transcript analysis view (frontend + backend processing)
- Post-speech performance breakdown (frontend display + backend analytics)

9. Technical Requirements (Backend Focus)

- Tech stack: Python backend using Django REST Framework (built-in admin panel supports the configurable requirements)
- Speech-to-text: free/open-source tool (e.g., OpenAI Whisper) — paid services are avoided; supports Bangla as well as English
- Filler-word detection: keyword/pattern-based; the word list is stored in the database and is configurable by admins (and optionally users), pre-loaded with an initial word list for demonstration
- WPM calculation from transcript + audio duration; pacing and pause thresholds kept configurable
- Pause detection from the gaps between words (the word times come from the speech engine)
- Recordings must be between 10 seconds and 5 minutes (admins can change both limits). The uploaded audio is deleted as soon as it has been processed. The transcript and the measurements are kept in the database so progress over time can be shown
- API endpoints for: prompt library, upload/recording, transcript retrieval, analysis results, filler-word list management
- Database schema for users, prompts, recordings, analysis results, and configurable filler words/settings

10. Success Metrics

- Accuracy of filler-word detection against manually reviewed samples
- WPM calculation matches manual count within an acceptable margin
- End-to-end latency from recording stop to feedback shown
- User retention / repeat rehearsal attempts per prompt

11. Implementation Status (Backend)

Built and tested

- Practice prompt library with categories, search and filters, managed in the admin panel
- Accounts: register, login with tokens, normal users and admins
- Recording upload with checks on format, size and length; processing runs in the background
- Speech to text with Whisper (faster-whisper), English and Bangla; a mock engine is used when Whisper is not ready, and the result says so
- Words per minute overall and over time, pace status against a target range
- Pause detection (short, dramatic, long) and filler-word detection with global and personal word lists
- Transcript, performance breakdown, recording history, progress over time and admin settings through the API
- A small test page for trying everything in the browser, and 136 automated tests

For the frontend group

- Recording interface, transcript view and performance breakdown screens, built on the API (the test page shows how the calls work)

Not built

- Pitch variation, open-ended question tracking, punchline timing and audience-type pacing mode (section 5.1)
- The camera-based features (section 5.2)
- Deployment: the system runs locally, there is no deployed link yet

Not measured yet

- The success metrics in section 10 need real recordings. Filler-detection accuracy, pace accuracy and processing time have not been measured with real speakers
- Whisper may leave out "um" and "uh", which would lower filler counts. The first test with real speech is still to be done and recorded in the project log
- Words like "like" and "so" are often used normally, so filler counts can be too high
- Bangla recognition is likely less accurate than English